0580_s22_ms_41_Q5

Question	Answer	Marks	Partial Marks
5(a)(i)(a)	$\frac{(8-2)\times 180}{8\times 2}$ oe	**M2**	**M1** for $\frac{(8-2)\times 180}{8}$ or $\frac{360}{8}$ or $\frac{(2\times 8-4)\times 90}{8}$
5(a)(i)(b)	174 or 173.8....	**4**	**M3** for $\frac{1}{2}\times 6\times OM$ oe or $\frac{1}{2}\times (OA)^2\times \sin 45$ oe or $\frac{1}{2}\times 6\times OA\times sin67.5$ oe where OA and OM are as in the **M2** or **M2** for $OM = 3\times \tan 67.5$ oe or for $OA = \left(\frac{3}{cos67.5}\right)$ or $\frac{6\times sin67.5}{sin45}$ oe or **M1** for $\frac{OM}{3} = \tan 67.5$ oe or for $\frac{3}{OA} = cos67.5$ oe or for $\frac{sin45}{6} = \frac{sin67.5}{OA}$ oe
5(a)(ii)	193 or 193.0 to 193.1	**3**	**M2** for $\pi\times\left(\frac{3}{cos67.5}\right)^2$ oe or **M1** for $\frac{3}{r} = cos67.5$ or $\frac{sin45}{6} = \frac{sin67.5}{r}$

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Question	Answer	Marks	Partial Marks
5(b)(i)	1.27 or 1.272 to 1.273	**2**	**M1** for $\left[\frac{1}{2}\times\right]\pi\times 0.45^2\times 4$ or $\frac{1}{2}\times\pi\times 0.45^2\left[\times 4\right]$
5(b)(ii)	742 or 743	**6**	**M5** for a method leading to the volume of water e.g. $4\times\{2\times\frac{invcos\left(\frac{0.15}{0.45}\right)}{360}\times\pi\times 0.45^2$ $-\frac{1}{2}\times 0.45^2\times\sin\left(2\,invcos\left(\frac{0.15}{0.45}\right)\right)\}$ oe **OR** **M2** $\left[2\times\right]\frac{invcos\left(\frac{0.15}{0.45}\right)}{360}\times\pi\times 0.45^2$ oe or $\left[2\times\right]\frac{90-invcos\left(\frac{0.15}{0.45}\right)}{360}\times\pi\times 0.45^2$ oe or **M1** for use of $\frac{\theta}{360}\times\pi\times 0.45^2$ oe **M2** for $\frac{1}{2}\times 0.45^2\times\sin\left(2\,invcos\left(\frac{0.15}{0.45}\right)\right)$ oe or $\frac{1}{2}\times 0.15\times 0.45\times\sin\left(invcos\left(\frac{0.15}{0.45}\right)\right)\left[\times 2\right]$ oe

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Question	Answer	Marks	Partial Marks
5(b)(ii)			**or M1 for use of** $\frac{1}{2}\times 0.45^2$ **× sinθ oe** **or** $[2\times]\frac{1}{2}\times 0.15\times 0.45\times \sin\beta$ **oe** **If 0 scored,** **SC1 for *inv*cos** $\left(\frac{0.15}{0.45}\right)$ **or *inv*sin** $\left(\frac{0.15}{0.45}\right)$ **or** $\sqrt{0.45^2-0.15^2}$ **soi**